

INWZA 007: Integrated Neural Weighted Zoom Architecture for Early Barrett’s Neoplasia Detection

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Code Availability: <https://github.com/snowoakk/RARE26-SadudeeP>

Team SadudeeP proposed INWZA 007 (Integrated Neural Weighted Zoom Architecture version 7), an efficient and robust framework designed to solve two major challenges in detecting early Barrett’s neoplasia. The first challenge is extreme class imbalance, meaning cancerous samples are incredibly rare compared to healthy tissue. The second challenge is domain shift, which refers to the visual differences in medical images because different hospitals use different cameras, lighting, and preparation methods. To solve these problems, our team structured the methodology into a clear, step-by-step pipeline (Figure 1). We first use an ensemble of preliminary mother models to automatically generate a robust, semi-supervised training corpus of highly reliable examples. We then use this enriched dataset for the rigorous training of our final child models. During deployment, these models are enhanced with advanced inference optimization and mathematical corrections to make highly accurate final predictions.

To capture the subtle visual details of cancer across a broad patient population, the framework first deploys an advanced semi-supervised pseudo-labeling pipeline Lee (2013) on the massive, unannotated GastroNet-5M dataset Jong et al. (2026). Because these millions of medical images lack human doctor labels, we initiate the process by training an ensemble of preliminary mother models: ConvNeXt-Base Liu et al. (2022), ConvNeXt-Tiny Liu et al. (2022), and EfficientNetV2 Tan and Le (2021). To counteract the extreme dataset imbalance and mathematically force the models to focus on exceptionally rare abnormalities rather than just common healthy tissue, they are optimized using a robust Focal Loss function ($\gamma = 2.0$). Once trained, these Mother models evaluate the unannotated data, blending their diagnostic insights via a strict weighted consensus: 50% for ConvNeXt-Base, 30% for ConvNeXt-Tiny, and 20% for EfficientNetV2. To guarantee the absolute quality of this newly generated data, stringent confidence thresholds are applied to isolate only the most reliable samples. Predictions scoring below 0.15 are confidently flagged as negative (healthy), while those above 0.85 are confidently flagged as positive (cancerous). These high-confidence pseudo-labels are then extracted to form a refined, enriched training corpus, acting as a high-quality foundation for the final student networks.

Leveraging the pseudo-labeled corpus, the original data, and newly generated synthetic cancer samples, created by extracting rare cancer shapes, altering them, and pasting them onto normal GastroNet images, our framework effectively transfers knowledge from the mother models into three powerful final child models. To establish a diverse understanding of the images, we integrated three distinct visual architectures: ConvNeXt Base Liu et al. (2022) initialized with DINOv3 weights Simeoni et al. (2025), Vision Transformer Base (ViT-Base) Dosovitskiy et al. (2021) initialized with DINOv2 weights Oquab et al. (2023), and ResNet-50 He et al. (2016) initialized with domain-specific GastroNet-5M weights Jong et al. (2026). Instead of basic output layers, each model is upgraded with a custom multi-layer projection classifier to better process complex image features. The DINOv3-ConvNeXt (512×512 resolution) condenses 1024-dimensional features into a binary output using batch normalization, ReLU activation, and 40% dropout. Meanwhile, the ResNet-50 (512×512 resolution) and DINOv2-ViT (518×518 resolution) process 2048- and 768-dimensional features, respectively, through two-stage projection heads (shrinking from 512 to 128 dimensions), stabilized by Layer Normalization, GELU activations, and 30% dropout. To guarantee these models learn the true medical signs of cancer rather than just memorizing different hospitals’ camera settings or lighting, training utilizes a leave-one-center-out cross-validation scheme alongside strict Exponential Moving Average (EMA) weight updates Morales-Brotons et al. (2024) (decay rate of 0.999) for stability. Finally, to ensure the models confidently highlight abnormalities, they are optimized using a specialized Surrogate PPV Ranking Loss that mathematically forces true cancer predictions to stand out clearly above normal background noise. For ResNet-50 and ViT, this combines a standard Cross-Entropy loss (weight=1.0) with a pairwise margin ranking loss (weight=2.0). The top-performing DINOv3-ConvNeXt child model takes this further by employing a scheduled learning approach called curriculum blending; it starts with weighted Cross-Entropy (0.05 label smoothing) and gradually shifts to a differentiable surrogate PPV loss (recall threshold=0.90, $\lambda = 50.0$) by stepping up the blend ratio (α) across progressive epoch steps.

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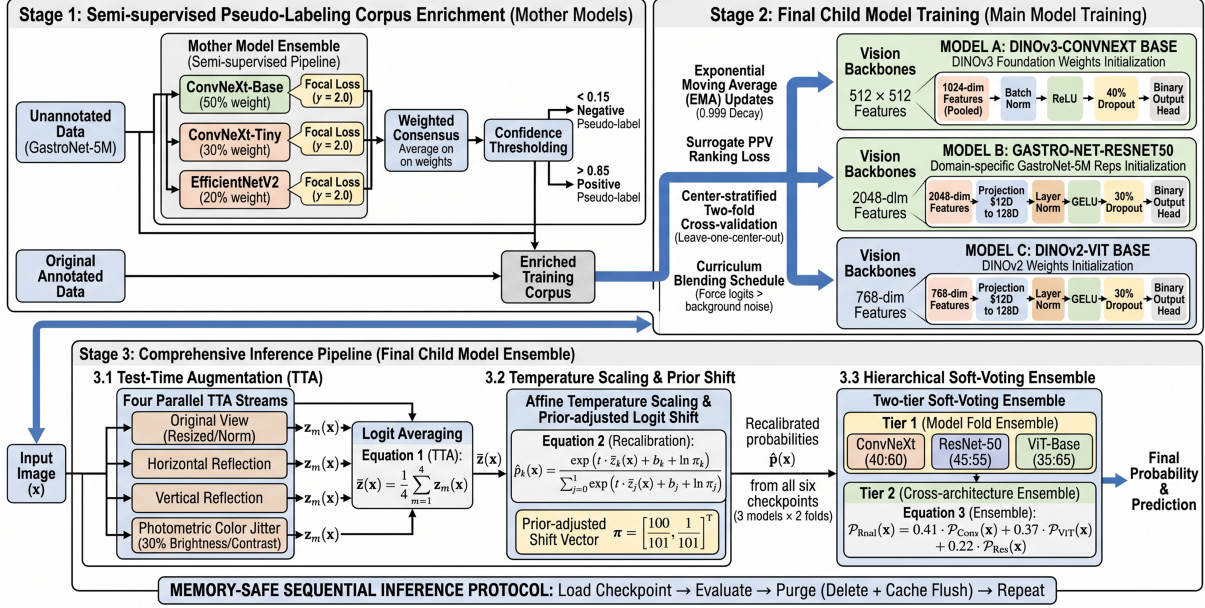


Figure 1: Overview of the Integrated Neural Weighted Zoom Architecture (INWZA 007). The pipeline illustrates the semi-supervised pseudo-labeling by the mother models, the multi-backbone student networks, test-time augmentation (TTA), logit calibration, and the final hierarchical soft-voting ensemble.

During inference, inputs undergo a robust test-time augmentation (TTA) pipeline to mitigate frame-level noise and motion artifacts. Each image processes through four parallel streams: standard resizing with ImageNet normalization, horizontal reflection, vertical reflection, and photometric color jittering with 30% brightness and contrast perturbation. For an input image $\mathbf{x}$, logit vectors $\mathbf{z}_m(\mathbf{x}) \in \mathbb{R}^2$ from all four views are averaged to produce a smoothed representation $\bar{\mathbf{z}}(\mathbf{x})$ (Equation 1):

$$\bar{\mathbf{z}}(\mathbf{x}) = \frac{1}{4} \sum_{m=1}^4 \mathbf{z}_m(\mathbf{x}) \quad (1)$$

To correct severe probability miscalibration induced by the 1:100 target-to-background class imbalance Godau et al. (2025), uncalibrated logits are adjusted using affine temperature scaling Guo et al. (2017) combined with a prior-adjusted logit shift Menon et al. (2021). A learned scalar temperature t and two-dimensional bias vector $\mathbf{b}$ are applied to the averaged logits, alongside an explicit prior probability shift vector $\boldsymbol{\pi} = [\frac{100}{101}, \frac{1}{101}]^T$ representing the clinical screening ratio. The recalibrated clinical probability $\hat{p}_k(\mathbf{x})$ is calculated by Equation 2:

$$\hat{p}_k(\mathbf{x}) = \frac{\exp(t \cdot \bar{z}_k(\mathbf{x}) + b_k + \ln \pi_k)}{\sum_{j=0}^1 \exp(t \cdot \bar{z}_j(\mathbf{x}) + b_j + \ln \pi_j)} \quad (2)$$

Predictions are finally integrated via a two-tier hierarchical soft-voting ensemble. Tier one blends fold-level predictions within each model family based on empirical validation ratios: ConvNeXt at 40:60, ResNet-50 at 45:55, and ViT-Base at 35:65. Tier two merges these cross-architecture predictions to capitalize on the unique strengths of each model family. We assign them different voting weights based on their out-of-center performance metrics, specifically, how well they generalized to outside hospital data. This results in final assigned weights of 41% for ConvNeXt Base, 37% for ViT Base, and 22% for ResNet-50. The final combined prediction $\mathcal{P}_{\text{final}}(\mathbf{x})$ is calculated by Equation (3):

$$\mathcal{P}_{\text{final}}(\mathbf{x}) = 0.41 \cdot \mathcal{P}_{\text{Conv}}(\mathbf{x}) + 0.37 \cdot \mathcal{P}_{\text{ViT}}(\mathbf{x}) + 0.22 \cdot \mathcal{P}_{\text{Res}}(\mathbf{x}) \quad (3)$$

The INWZA 007 ensemble achieved remarkable results in the hidden test set during the Open Development Phase, delivering an outstanding AUROC of 0.9098 and a positive predictive value (PPV) of 0.0335 at 90% recall. This high AUROC score validates the strong discriminative capacity of the model to correctly rank and order neoplastic cases relative to normal controls, demonstrating excellent generalization across different clinical centers. Beyond these metrics, a major achievement of this work is the innovative methodology that integrates semisupervised learning with an advanced ensemble framework to create unprecedented computational efficiency. Deploying a massive pipeline of six deep neural network checkpoints coupled with test time augmentation across

four views naturally introduces severe GPU memory and latency demands that typically prevent parallel batch execution. To bypass these strict hardware limits, our team engineered a novel memory safe sequential inference protocol. Each checkpoint is loaded sequentially, evaluated, and immediately purged using explicit object deletion and CUDA cache flushing. This strategic pipeline completely prevents memory exhaustion faults while maintaining a highly efficient average processing speed of 0.04 seconds per image. This breakthrough proves that heavy multiple architecture ensembles can be safely and efficiently deployed in clinics with limited hardware resources if resource purging is strictly managed.

Despite these significant achievements, primary limitations remain regarding absolute predictive precision and domain generalizability. While the model achieves a highly competitive AUROC and a strong PPV relative to the benchmark, the absolute PPV of 0.0335 illustrates a persistent barrier to clinical deployment. Under an extreme 1:100 target prevalence, the system still generates a high volume of false positive alarms. Mitigating this issue requires further refinement of calibration techniques. Additionally, because the architecture relies on data from only two medical institutions, it remains vulnerable to site specific overfitting. To establish robust generalizability, future iterations will adopt a hierarchical cross validation strategy by partitioning the data by center, performing internal k-fold training and testing strictly within each individual center, and evaluating these internal metrics before conducting final evaluations across centers. Finally, future enhancements will explore targeted hard negative mining during the pseudo labeling loops and nonparametric post hoc recalibration. These targeted steps are designed to aggressively suppress false positive rates and refine absolute predictive precision in automated screening pipelines.

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